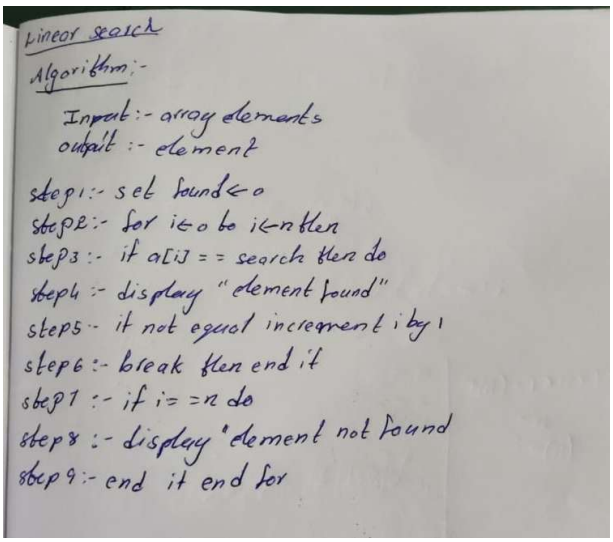
PRACTICAL: 02

AIM :- Implementation and Time analysis of linear and binary search Algorithm.

Algorithms:- linear search

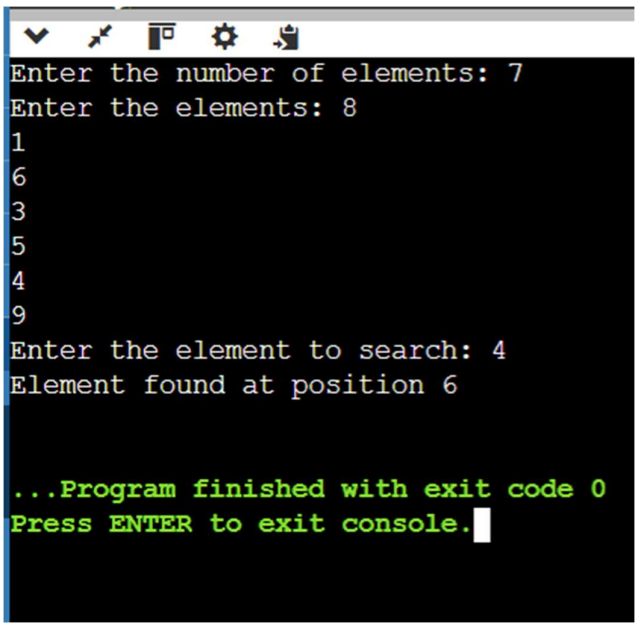


Program/code:-

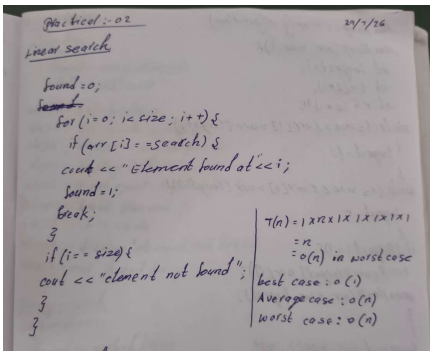
```
#include <iostream>
using namespace std;
int main() {
    int n, key;
    cout << "Enter the number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter the elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
```

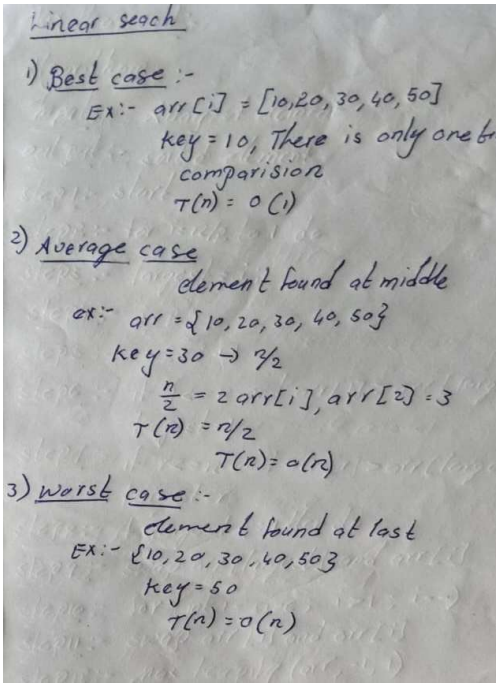
```
    cout << "Enter the element to search: ";
    cin >> key;
    int found = 0;
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            cout << "Element found at position " << i + 1 << endl;
            found = 1;
            break;
        }
    }
    if (found == 0)
        cout << "Element not found." << endl;
    return 0;
}
```

Output:-

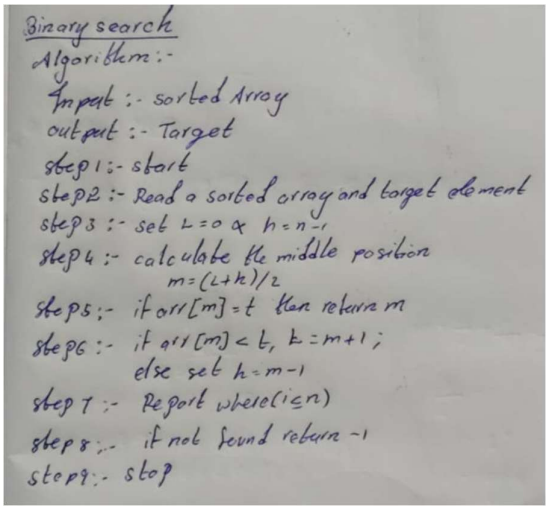


Time complexity:-





Algorithms:- binary search

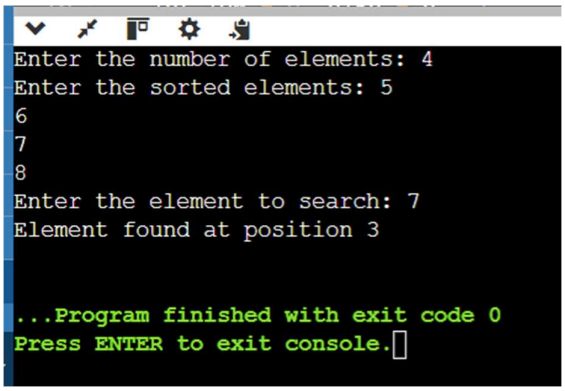


Program/code:-

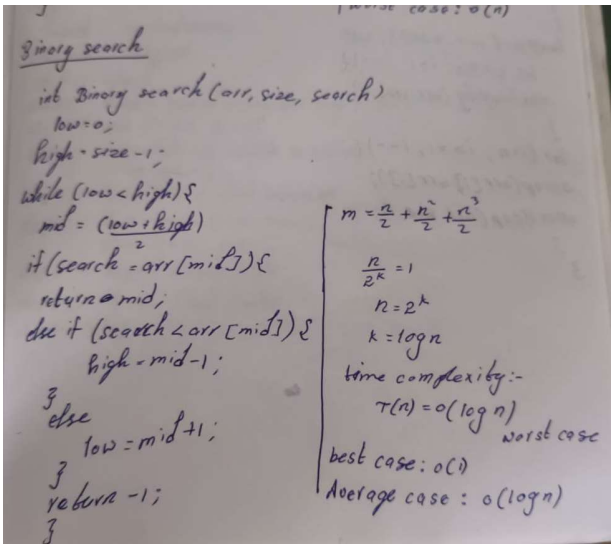
```
#include <iostream>
using namespace std;
int main() {
    int n, key;
    cout << "Enter the number of elements: ";
    cin >> n;
    int arr[n];
    cout << "Enter the sorted elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "Enter the element to search: ";
```

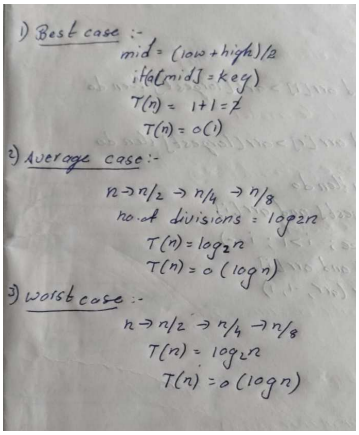
```
    cin >> key;
    int low = 0, high = n - 1;
    int found = 0;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            cout << "Element found at position " << mid + 1 << endl;
            found = 1;
            break;
        }
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    if (found == 0)
        cout << "Element not found." << endl;
    return 0;
}
```

Output:-



Timecomplexity:-





Conclusion:-

In conclusion, Linear Search and Binary Search are important searching algorithms. Linear Search is simple and can be used on both sorted and unsorted data, but it takes O(n) time in the worst case. Binary Search is much faster with O(log n) time, but it requires the data to be sorted.

Therefore, Linear Search is suitable for small or unsorted datasets, while Binary Search is preferred for large sorted datasets because of its better efficiency